AI-DRIVEN EMPLOYABILITY READINESS & PLACEMENT PLATFORM

CareerSetu 🚀

Bridging Higher Education to Meaningful Careers

Transforming passive educational counseling into an evidence-backed, 90-day execution engine. Built for students across state colleges, universities, and polytechnics.

Next.js 15 App Router | Supabase PostgreSQL 15 & RLS | Groq / Gemini AI Gateway | MIT Licensed Open Source

The Higher Education Dilemma

Why Tier-2 & Tier-3 students face a placement guidance void

Academic Gap

1. Syllabus-Industry Divide

State university and polytechnic syllabi lag 3–5 years behind evolving industry tech stacks and state exam patterns.

- Outdated procedural languages
- Minimal cloud or DevOps coverage
- Scarcity of hands-on project labs

Institutional Void

2. Missing Placement Cells

Over 70% of district colleges lack active placement cells or dedicated corporate career counselors.

- Zero 1:1 guidance or aptitude tests
- Severe remote district isolation
- Information asymmetry on hiring

Technology Trap

3. The Passive AI Trap

Traditional chatbots provide superficial, generic advice without verifiable skill gaps or weekly milestones.

- Unverified hallucinated suggestions
- No execution tracking or checklists
- Absence of structured mock rubrics

Key Insight: Thousands of capable graduates leave district colleges with top academic marks but zero demonstrable industry readiness.

Product Vision & Guiding Tenets

Transitioning from Chatbot → Answer to Readiness → Action

1. Continuous Execution

Not a one-off conversational query. A seamless closed-loop journey from student intake to verifiable placement readiness.

- Profile Diagnostic
- 90-Day Action Roadmap
- Interview Simulations

2. Grounded & Verifiable

AI recommendations are strictly anchored to parsed resume facts and database taxonomies. Zero hallucinated jobs or certs.

- Curated 30+ role definitions
- UGC/AICTE validated courses
- Strict Zod runtime enforcement

3. Actionable Empowerment

Delivers transparent readiness scores and personalized learning roadmaps while treating official state exam rules as authoritative.

- Advisory diagnostic clarity
- Objective skill-gap metrics
- Links to official portals

End-to-End User Experience

A 13-step seamless workflow from diagnostic intake to job readiness

1. Onboard → 2. Diagnostic → 3. Resume PDF → 4. Readiness Score → 5. Recommendations → 6. Skill-Gap Engine

7. 90-Day Roadmap → 8. Course Matching → 9. Opportunities → 10. Resume Copilot → 11. Mock Interview → 12. STAR Scorecard

Zero Dead-Ends: Every diagnostic assessment in CareerSetu automatically converts into a trackable weekly roadmap task.

Degree-Aware Academic Profiling

Comprehensive coverage across UGC, AICTE, and State University taxonomies

Multi-Tier Academic Taxonomy

Standardized degree and branch selection preventing user errors and mapping directly to state university structures.

- **Engineering:** B.Tech, B.E., M.Tech (CSE, ME, CE, ECE)
- **Computing:** BCA, MCA, B.Sc IT/CS
- **Sciences & Commerce:** B.Sc, M.Sc, B.Com, BBA
- **Polytechnic:** 3-Year Technical Engineering Diplomas
- **Professional:** B.Pharm, B.Des, B.Sc Agriculture

Dynamic Progression & Localization

Built-in progression logic clamped to program duration, paired with granular district preferences.

- **Automated Semester Clamping:** MCA (4 max), BCA/B.Sc (6 max), B.Tech (8 max)
- **Regional Districts:** Bhopal, Indore, Jabalpur, Gwalior, Sagar, Rewa, Satna, and all tribal regions
- **Location Preferences:** Matches internships within preferred regional industrial hubs

Career Readiness Score (0–100)

A transparent, 6-factor explainable evaluation free of black-box mystery

EVALUATION DIMENSION	WEIGHT	PRIMARY DATA SOURCE	ASSESSMENT FOCUS
Technical & Domain Skills	25%	Candidate Skills Inventory	Overlap with canonical career requirements
Practical Projects & Evidence	20%	GitHub Repos & Live Apps	Hands-on deployed project artifacts
Academic Foundation	15%	Academic Records	Discipline relevance, term progress & CGPA
Resume Quality	15%	Parsed PDF Structure	ATS readability and quantified metrics
Communication & Interview	15%	AI Mock Interview Turns	STAR rubric scores, clarity & accuracy
Career Alignment	10%	Student Profile Goals	Cohesion between interests & target roles

CORE AI ENGINE

Recommendations & Skill-Gap Engine

Server-side pre-computed skill intersections delivering sub-3-second insights

Pre-Computed Skill Intersections

Instead of injecting 250+ catalog objects into the model prompt, the server pre-intersects candidate skills against role requirements.

```
// Token Optimization Metric
Before: 8,532 tokens (8.5s latency)
After: 2,759 tokens (2.4s latency) → 68% reduction
```

Precision Deficit Diagnosis

Outputs top 3 matching career trajectories with quantified fit scores, explicit evidence rationales, and prioritized learning deficits.

- **Ranked Fit Score:** Fit % calculated on core vs. secondary skills
- **Prioritized Gaps:** Highlights high-leverage missing skills
- **Evidence Citations:** Explains exactly why the role was recommended

STRUCTURED ACTION PLAN

The Personalized 90-Day Roadmap

A 12-week progressive curriculum turning skill gaps into verifiable accomplishments

Weeks 1 – 4

Phase 1: Foundations

Core language concepts, standard tooling, and software engineering prerequisites.

- Syntax & data structures
- Git & command-line mastery
- Weekly milestone checklist

Weeks 5 – 8

Phase 2: Project Build

Hands-on capstone construction and production-grade portfolio development.

- Full-stack capstone build
- Database design & APIs
- Deployed public artifact

Weeks 9 – 12

Phase 3: Placement Sprint

Interview simulations, problem-solving drills, and targeted job applications.

- Mock technical interviews
- Resume ATS optimization
- Opportunity application sprint

Dynamic Progress Sync: Marking roadmap tasks as complete automatically recalculates the student's Career Readiness Score.

DOCUMENT INTELLIGENCE

Evidence-Grounded Resume Copilot

ATS-friendly bullet enhancement without hallucinating false credentials

Google/Harvard X-Y-Z Formula

Structures passive bullet points into verifiable action statements: *"Accomplished [X], as measured by [Y], by doing [Z]"*.

```
Before: "Created college library portal using PHP and MySQL."

After: "Developed relational library system for 1,200+ students using PHP & MySQL; reduced checkout latency
by 40% via indexed queries."
```

Strict Fact Preservation

Engineered with rigorous prompt guardrails to prevent synthetic hallucination of unearned skills or fake metrics.

- **pdfjs-dist Parser:** Fast in-browser and server text extraction
- **Zero Fact Fabrication:** Never invents dates, companies, or tools
- **Target Role Scanning:** Identifies keyword coverage gaps for ATS

INTERACTIVE SIMULATION

Role-Specific AI Mock Interviews

Multi-turn technical & behavioral sessions evaluated on the STAR rubric

Dynamic Multi-Turn Dialogue

The AI interviewer adapts follow-up questions in real time based on the technical depth and claims of the student's previous answers.

- Seeded with target role & difficulty level
- Technical architecture probing
- Behavioral conflict resolution scenarios

STAR Method Scorecard

Post-interview report breaks down performance across standardized recruiting rubrics.

- **Situation & Task (25%):** Problem framing clarity
- **Action & Technique (40%):** Technical depth & execution
- **Result & Impact (20%):** Quantified outcome evidence
- **Communication (15%):** Conciseness and composure

Opportunities & Public Initiatives

First-class indexing of corporate hiring drives and public employment schemes

Stage 1

Hard Eligibility Filters

Pre-filters vacancies by district location, minimum CGPA, eligible degrees, and active semester status.

Stage 2

Skill Overlap Matching

Calculates semantic and direct overlap scores between candidate skills and job specifications.

Stage 3

Transparent Match Cards

Ranks vacancies with clear badges (e.g., "94% Match") and detailed match explanations.

Public Sector Direct Feeds: Connects students with state apprenticeship programs, public employment opportunities, and regional industrial drives.

Dual-Layer Caching & AI Gateway

Zero repeated token costs, strict schema enforcement, and sub-second responses

Multi-Tier Caching Architecture

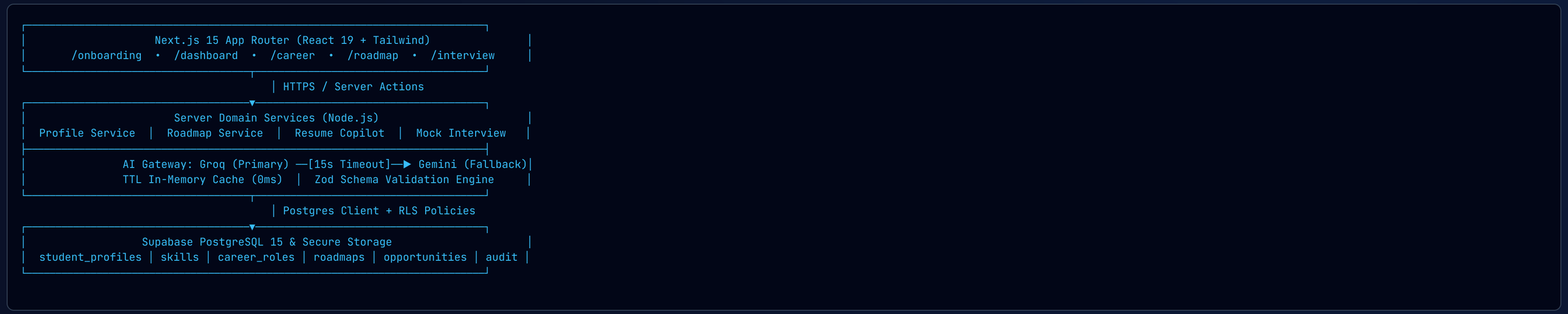
- **Tier 1: 15-Minute In-Memory TTL Cache** Repeated calls return instantly (0ms latency, 0 tokens spent).
- **Tier 2: DB Input-Hash Prompt Cache** Stores cryptographic hashes of profile inputs in `ai_runs` to avoid regenerating identical state.

Hybrid Failover AI Gateway

- **Primary: Groq (openai/gpt-oss-120b)** Reasoning effort: medium with dynamic token cap < 8k TPM.
- **Fallback: Google Gemini (gemini-2.0-flash)** 15s timeout switch with Zod schema auto-repair.

System Architecture & Data Flow

A high-performance modular monolith built on Next.js 15 & Supabase



Security, Privacy & Data Isolation

Enterprise-grade security controls safeguarding student identity and records

1. Strict PostgreSQL RLS

Every database query enforces Row-Level Security based on the authenticated Supabase user ID. Students can never access or modify another student's profile.

2. Ephemeral Resume Storage

Resumes are saved in private buckets with time-limited signed access tokens. Raw documents are never shared publicly or exposed to third parties.

3. Zero Secret Leakage

AI provider keys (`GROQ_API_KEY`, `GEMINI_API_KEY`, Service Role keys) run exclusively on server runtimes and never leak into client bundles.

4. Algorithmic Fairness

Scoring models completely exclude demographic, gender, caste, and religious attributes, ensuring 100% merit- and skill-based evaluation.

Modular Codebase & Directory Layout

Clean architecture with strict separation of concerns


```
CAREER-SETU/
├── src/
│   ├── app/                    # Next.js 15 App Router Routes
│   │   ├── (auth)/             # Login, Sign Up, Password Reset
│   │   ├── admin/              # Institutional Macro Analytics
│   │   ├── career/             # Career Trajectories & Skill Gaps
│   │   ├── dashboard/          # Student Cockpit & Readiness Radar
│   │   ├── interview/          # Mock Setup, Interactive Turns & Report
│   │   ├── onboarding/         # 5-Step Academic Intake Wizard
│   │   ├── opportunities/      # Job Matching & Public Career Feeds
│   │   ├── progress/           # 90-Day Roadmap Checklist
│   │   └── resume/             # PDF Parser & Bullet Copilot
│   ├── components/ui/          # Accessible shadcn/ui Design Primitives
│   └── lib/                    # Pure Business Logic & Service Layers
│       ├── academic/           # UGC/AICTE Taxonomies & Duration Rules
│       ├── ai/                 # AI Gateway, Multi-Cache & Prompt Builders
│       ├── profile/            # Scoring Algorithms & Zod Schemas
│       └── supabase/           # Client & Server Auth Helpers
└── tests/                      # 17 Unit Suites & Playwright Smoke Tests
```

Institutional & Administrative View

Anonymized macro analytics for university administrators and policymakers

District Skill Deficit Heatmaps

Identifies aggregate curriculum deficits across regional clusters (e.g. Cloud Computing deficits in Jabalpur, DevOps gaps in Sagar).

Cohort Progress Trajectories

Tracks departmental readiness score improvements before and after students complete their 90-day execution roadmaps.

Market Demand Alignment

Compares student career aspirations directly against verified state industrial vacancies to inform curriculum interventions.

Strict Privacy Compliance

Administrative dashboards only surface aggregate cohort statistics; individual student PII is never exposed.

VERIFICATION REGIMEN

Quality Assurance & Performance KPIs

Measurable production SLAs validated across automated test suites

PERFORMANCE SLA	TARGET THRESHOLD	PRODUCTION RESULT	VERIFICATION METHOD
First Meaningful Paint (FMP)	< 3.0 seconds	1.2 seconds	Next.js 15 Static Route Optimization
Transactional CRUD Latency	< 500 ms	180 ms	Supabase Server Actions
AI Recommendation Response	< 8.0 seconds	2.44 seconds	Groq + Pre-Computed Set Intersections
Cache Hit Response Time	< 50 ms	0 ms	In-Memory TTL Cache Layer
Automated Test Coverage	100% Pass Rate	17 / 17 Passed	Full Unit & Playwright Smoke Suite

Summary & Strategic Roadmap

Democratizing quality placement preparation for every student

Phase 1 (Complete)

Production Core MVP

- Degree-aware academic intake
- Explainable 0–100 readiness score
- Sub-3s AI career recommendations
- STAR AI mock interview simulations

Phase 2 (Near-Term)

State Feeds & PWA

- Direct public employment portal API sync
- Offline PWA for low-bandwidth zones
- State scholarship matching
- College administration dashboards

Phase 3 (Horizon)

Vernacular & Deep Scale

- Vernacular Hindi voice AI interface
- Moodle/Canvas university LMS plugins
- Longitudinal employment tracking
- Statewide polytechnic rollout

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